

Q1: NOV 2001 P2

The unit of a physical quantity may be shown with a prefix. For example, the prefix *micro* (μ) has the decimal equivalent 10^{-6} , so that 1 microamp ($1\mu\text{A}$) can be written as 10^{-6}A . Complete Fig. 3.1 to show each prefix with its corresponding decimal equivalent.

prefix	decimal equivalent
pico	
micro	10^{-6}
giga	
.....	10^{12}

Fig. 3.1**[3]****Q2: NOV 2002 P2**

- 1 (a) The following equations may or may not be correct. Use base units to check if they represent the given relationships.

- (i) The fundamental frequency f , of an open pipe of length L , when the speed of sound is v , $f = \frac{v}{2L}$.

- (ii) The pressure p at a point in a fluid of density ρ moving with a speed v , $p = \frac{1}{2}\rho v$.

- (iii) The angle θ at which an aeroplane moving with a speed v should be banked to describe a path of radius of curvature r ,

$$\tan\theta = \frac{v^2}{rg}.$$

[6]

- (b) Torque and energy have the same dimensions.

- (i) Explain why it would be inappropriate to measure torque in joules.

- (ii) Write down an appropriate unit of torque.

[2]

Q3:JUN 2003 P2

- 1 Complete Fig. 1.1 to show each quantity and its unit.

[4]

quantity	unit
speed	m s^{-1}
density	
.....	s^{-1}
electric field strength	
.....	kg m s^{-1}

Fig. 1.1

Q4:2006 SPEC P2

1 (a) Express each of the following units in terms of base units.

(i) the joule

(ii) the pascal

(iii) the volt

4
[6]

(b) Distinguish between vector and scalar quantities. Give an example for each.

[3]

Q5:NOV 2007 P2

1 (a) State any **three** base units.

[3]

1.0 GENERAL PHYSICS P2&P3 QUESTIONS

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- (b) The focal length f , of a lens is given by

$$f = \frac{uv}{u + v},$$

where u is the object distance and v is the image distance. In one particular situation, values of u and v were

$$\begin{aligned} u &= (23.1 \pm 0.1) \text{ cm} \\ v &= (18.20 \pm 0.01) \text{ cm.} \end{aligned}$$

Determine the uncertainty in f and hence f .

uncertainty in $f =$ _____

$f =$ _____ $\pm$ _____

[5]

Q6:NOV 2009 P2

- 1 Complete the table to distinguish between a *systematic error* and a *random error*.

	systematic	random
cause		
example		
effect		
treatment		

[8]

Q7: JUN 2010 P2

- 1 (a) A student measuring the diameter of a piece of wire mistakenly used a micrometer screw gauge which did not read zero when fully closed.

- (i) State with a reason if her readings have a random error or a systematic error.

- (ii) Explain how this error can be reduced.

[3]

- (b) A test tube has the following diameters:

external diameter $d_1 = (17,9 \pm 0,5)$ mm

internal diameter $d_2 = (13,6 \pm 0,5)$ mm

Determine the percentage uncertainty in $d_2 - d_1$.

= _____ [2]

Q8: JUN 2011 P2

- 1 (a) Distinguish between a *random error* and a *systematic error*.

[2]

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- (b) A student wishes to use a micrometer screw gauge to measure the diameter of a wire.

(i) Suggest how the student can

1. reduce the systematic error in a reading,

2. allow for a non-circular cross section of the wire,

3. allow for a wire of varying diameters.

- (ii) The volume, V , of a cylinder of length, L , and radius, R , is given by

$$V = \pi R^2 L.$$

Calculate the radius of the cylinder and its uncertainty, given that

$$\begin{aligned} V &= (25.0 \pm 0.3) \text{ m}^3 \\ L &= (20.0 \pm 0.1) \text{ cm.} \end{aligned}$$

$$\text{radius} = \text{_____} \pm \text{_____} \quad [6]$$

Q9:NOV 2011 P2

- 1 (a) Use base units to check the homogeneity of the following equations.

(i) $T = \frac{1}{2\pi} \sqrt{\frac{l}{g}}$ where g is acceleration due to gravity, l the length and T the period.

(ii) $F = mv^2/r$ where v is the speed, m the mass, r the radius and F the centripetal force.

[4]

- (b) The equation $F = k\rho r^2 v^2$ where k is a constant relates to the resisting force, F , acting on a sphere of radius, r , moving with constant velocity, v , through a fluid of density, ρ .

Find the base units of the constant k .

base units of $k =$ _____ [3]

Q10: NOV 2012 P2

1 Express each of the following units in terms of base units.

(i) *volt*

(ii) *ohm*

(iii) *watt*

[6]

Q11: NOV 2013 P2

1 (a) A student stated "base units are used to check if an equation is correct".

(i) Define the term *base unit*.

(ii) State with a reason if the statement is true.

[3]

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- (b) A wire of length (1.94 ± 0.02) m and diameter (0.45 ± 0.02) mm has resistance $(25.1 \pm 0.1) \Omega$.

Calculate the resistivity, ρ , of the wire and its uncertainty.

$$\rho = \underline{\hspace{2cm}} \pm \underline{\hspace{2cm}} \quad [5]$$

Q12:JUN 2014 P2

- 1 (a) Define the following terms:

- (i) *derived quantity*

- (ii) *physical quantity*

[2]

- (b) The equation of state for one mole of a real gas is

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT,$$

where P , V and T are the pressure, volume and absolute temperature of the gas respectively.

- (i) Determine the base units of

1. a ,

base units of a = _____

2. R .

base units of R = _____

- (ii) State any two cases where an equation can be homogeneous but incorrect.

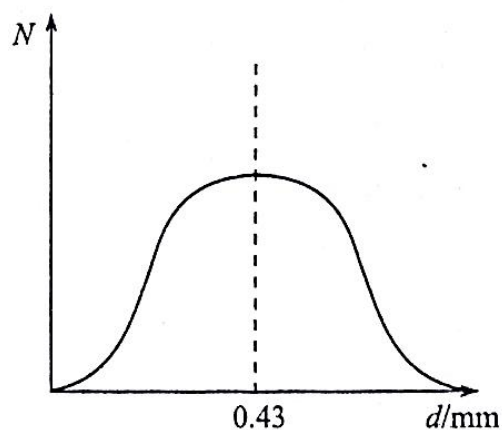
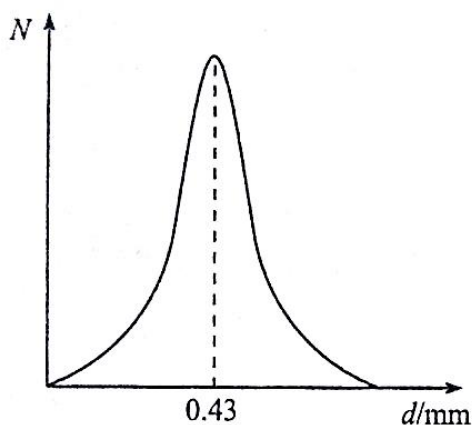
[5]

Q13:JUN 2015 P2

- 1 (a) (i) Distinguish between *precision* and *accuracy*.

 _____ [2]

- (ii) Fig.1.1 and Fig 1.2 show the distribution of measurements taken, for a constantan wire of diameter 0.43 ± 0.01 mm by two students.



Comment on the precision and accuracy of the measurements.

Fig.1.1 _____

Fig.1.2 _____

[4]

- (b) Give one advantage of repeated measurements over single measurements.

 _____ [1]

Q14: NOV 2015 P2

- (a) Distinguish between *precision* and *accuracy*.

- (b) The speed, V , of ocean waves is given by $V = k\sqrt{g\lambda}$, where k is a constant, λ is the wavelength and g is acceleration of free fall.

Determine the base units for k .

base units for k = _____ [2]

- (c) In a simple pendulum experiment to determine the period, T , the equation used is $T = 2\pi\sqrt{\frac{l}{g}}$ where T is found to be (2.16 ± 0.01) s and the length, l , of pendulum is (1.15 ± 0.005) m.

Calculate the value of g and its uncertainty.

g = _____ $\pm$ _____ [3]

Q15: JUN 2016 P2

- 1 (a) Define the term *base unit*.

[1]

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- (b) The drag force, D , on an object of cross-sectional area, A , moving with speed, v , through a fluid of density, ρ , is given by

$$D = \frac{1}{2} C \rho A v^2,$$

where C is a drag constant.

- (i) Show that C has no units.

- (ii) The following measurements were obtained for a liquid of constant density 990 kgm^{-3} and drag constant 0.30.

$$A = 12.1 \pm 0.3 \text{ cm}^2$$

$$v = 10.0 \pm 0.2 \text{ ms}^{-1}$$

Calculate the drag force and its uncertainty.

drag force = _____ $\pm$ _____ N [6]

Q16:NOV 2016 P2

- 1 (a) Distinguish between a *vector quantity* and a *scalar quantity*.

- (b) Fig. 1.1 shows a system of forces acting at a point.

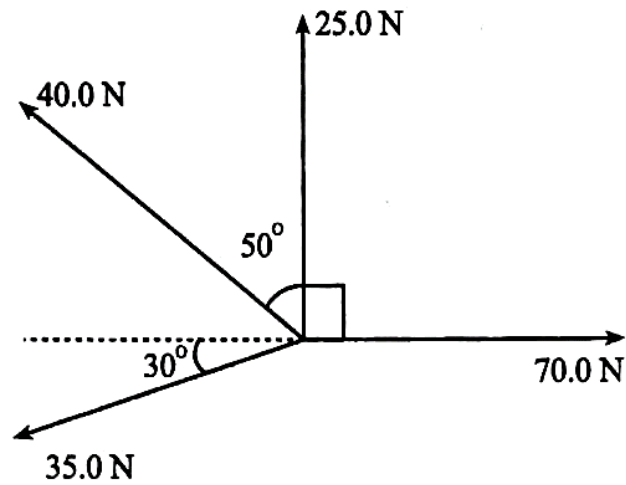


Fig. 1.1

Calculate the magnitude of the resultant force.

magnitude of resultant force = _____ [3]

Q17:JUN 2017 P2

- 1 (a) Explain how an accurate value of the diameter of a thin wire is measured.

[2]

- (b) The diameter, d , and length l , of a piece of wire are given as:

$$\begin{aligned} d &= (0.84 \pm 0.04) \text{ mm} \\ l &= (755 \pm 4) \text{ mm} \end{aligned}$$

The volume, V , of the wire is given by $V = \frac{1}{4} \lambda d^2 l$, where λ is a constant.

Calculate the percentage uncertainty in the measurement of the

- (i) diameter,

uncertainty = _____ %

- (ii) length,

uncertainty = _____ %

- (iii) volume.

uncertainty = _____ %

[4]

Q18:NOV 2018 P2

- 1 (a) Define the term *density*.

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- (b) A cylindrical coin has a mass of (12.6 ± 0.5) g, thickness of (2.00 ± 0.01) mm and a diameter of (21.2 ± 0.1) mm.

Calculate the density of the coin and its uncertainty.

density $\rho =$ _____

Q19:2018 SPEC P2

- 1 (a) Define *angular velocity*.

_____ [1]

- (b) A body is moving in a circular path of radius, r , at constant linear velocity, v , angular velocity, ω , and period T .

Deduce an expression connecting v , r and T .

_____ [1]

1.0 GENERAL PHYSICS P2&P3 QUESTIONS

Compiled by: R. Muga

Q20: NOV 2007 P3

- (1) (a) (i) Distinguish between *scalar* and *vector* quantities.
- (ii) Give an example of each. [4]

- (b) The resonance frequency, f , of a monometer wire is given by the equation

$$f = \frac{1}{2L} \sqrt{\left(\frac{4T}{\pi d^2 \rho}\right)} \text{ where}$$

T is the tension in the wire, L is the length of the wire, ρ is the density of the wire and d is the diameter of the wire.

- (i) Use base units to show that the equation is homogeneous.
- (ii) A wire of length (85.60 ± 0.01) cm, diameter (1.00 ± 0.01) mm and a constant density of 8800 kgm^{-3} resonates when the tension in the wire is (52.00 ± 0.05) N.
- Calculate the resonance frequency.
 - Calculate the uncertainty in the frequency.
 - Give your value for the frequency with its uncertainty quoted. [8]

- (c) (i) Distinguish between *systematic* and *random* errors.
- (ii) Give **two** examples of systematic errors and explain how each can be avoided.
- (iii) Discuss **two** ways in which random errors can be minimized. [8]

Q21: JUN 2008 P3

- 1 (a) The gravitational potential, V , at a point a distance, r , from the centre of the Earth, is given by

$$V = -\frac{GM}{r}, \text{ where } G \text{ and } M \text{ have their usual meanings.}$$

- (i) Explain the significance of the negative sign.
- (ii) Show that the equation is homogeneous. [3]

1.0 GENERAL PHYSICS P2&P3 QUESTIONS

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Q22: NOV 2008 P3

- 1 (a) In an experiment to determine the acceleration due to gravity g , the following results were obtained in ms^{-2} :
- 7.88; 7.86; 7.87; 7.86; 7.87.
- Comment on the precision and accuracy of these results.
 - Estimate the percentage uncertainty in the measurement of g .
 - Suggest the type of error associated with the results. [7]

Q23: JUN 2011 P3

- 1 (a) (i) Name any **two** physical quantities and their S.I. units.
- (ii) State any **two** properties that can be used to describe a physical quantity.
- (iii) Give **one** advantage of the S.I. system of units.
- (iv) Explain why the homogeneity of a physical equation is not sufficient for the equation to be correct.
- State how the correctness of the equation can be verified.
- (v) Describe how a reading can be **precise** but not **accurate**. [7]

Q24: JUN 2012 P3

- 1 (a) (i) List **six** base quantities and their units.
- (ii) Distinguish between *derived* and *base* units.
- (iii) Express the units of *coefficient of viscosity* in terms of base units. [7]
- 2 (a) (i) Give **one** example of a *systematic error*.
- (ii) State how the error in (i) can be minimised. [2]
- (b) The electric potential, V , is given by the formula, $V = \frac{Q}{4\pi\epsilon_0 r}$.
- Values of Q and r are $(3.2 \pm 0.1) \times 10^{-16} \text{ C}$ and $(1.3 \pm 0.02) \times 10^{-17} \text{ m}$ respectively.
- Calculate the error in the electric potential. [2]
- (c) (i) Explain the term *vector quantity*.

1.0 GENERAL PHYSICS P2&P3 QUESTIONS

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- 4 (a) Two lengths were recorded as (1.873 ± 0.005) mm and (1.582 ± 0.005) mm.
Calculate the fractional uncertainty in the sum and the difference of the lengths. [3]
- (b) Explain how a cathode ray oscilloscope can be used to measure time-interval. [3]

Q25:JUN 2013 P3

- 1 (a) (i) Give any **two** distinctions between random and systematic errors.
- (ii) A student used a micrometer screw-gauge to measure the diameter of a thin wire. He did not recognise that the reading was not zero when the gauge was fully closed.
- Outline how the random error in the measurement of the diameter of a thin wire is reduced.
 - Explain why the readings obtained by the student may be precise but not accurate.
- [5]

Q26:JUN 2014 P3

- 1 (a) (i) Define the terms *systematic error* and *random error*.
- (ii) Suggest **one** way of reducing each type of the errors in (i).
- (iii) When performing an experiment to determine a constant, E , of a material a student recorded the following values:

length of wire, l	$= (3.255 \pm 0.005)$ m
diameter of wire, d	$= (0.63 \pm 0.02)$ mm
force applied to wire, F	$= (26.5 \pm 0.1)$ N
extension of wire, e	$= (1.40 \pm 0.05)$ mm

The constant, E , of the wire is related to the measured values above by the relation

$$E = \frac{Fl}{Ae}$$

where A is cross-sectional area of the wire.

Calculate the fractional uncertainty in the value of E .

[6]

Q27: NOV 2014 P3

- 1 (a) (i) Express the farad in base units.
- (ii) The braking force of a vehicle is given by the equation.

$$F = x^2 A + \frac{P}{V}$$

where

x is the distance moved after applying the brakes

A is a constant

P is the power

V is the speed when brakes are applied

- Determine the base units for A .
- Suggest the identity of physical quantity A .

[5]

Q28: JUN 2015 P3

- 1 (a) (i) A force-extension graph for a spring which obeys Hooke's law was drawn and it was observed that:
- the plotted points did not follow a straight line
 - the intercept was not zero
- State the error associated with each observation.
- (ii) Explain whether or not a systematic error can be eliminated by averaging repeated readings.

[5]

Q29: NOV 2015 P3

- 1 (a) (i) State the base unit of temperature.
- (ii) Boltzmann's constant, k , and Planck's constant, h , are related to Stefan constant, δ , by the expression

$$\delta = \frac{ak^4}{h^3 c^2},$$

where a is a dimensionless constant and c is the speed of light.

Determine the base units of δ .

- (iii) State the limitation of using base units in the equation in (ii).

[5]

- (b) (i) Define the terms *accuracy* and *precision*.
- (ii) State and explain the error associated with precision.
- (iii) The following results in Nkg^{-1} were obtained in an experiment to determine the earth's gravitational field strength, g :

8.6; 8.7; 8.8; 8.5; 8.2; 8.9

Discuss whether the results were precise and/or accurate.

[6]

Q30: JUN 2016 P3

- (b) (i) Write down the decimal equivalent of the

1. pico,

2. deci.

- (ii) The pressure, P , of an ideal gas is given by the equation

$$P = \frac{1}{3} \rho \langle c^2 \rangle$$

where ρ is the density and $\langle c^2 \rangle$ is the mean square speed of the gas.

Use base units to check homogeneity of the equation.

[5]

- (c) (i) Explain the terms *accuracy* and *precision*.

- (ii) Explain why precise readings are not necessarily accurate readings.

[4]

Q31: NOV 2016 P3

- 1 (a) Fig. 1.1 shows a vernier callipers before and after making measurements of the internal diameter of a cylindrical tube. The tube has a length of 25.0 cm as measured by a ruler with an accuracy of 1 mm.

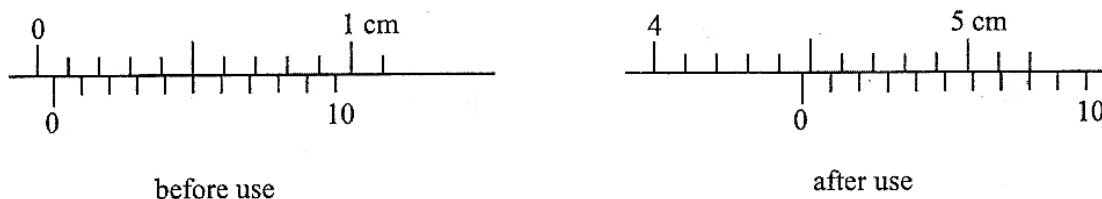


Fig. 1.1

1.0 GENERAL PHYSICS P2&P3 QUESTIONS

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- (i) Write the readings shown in Fig. 1.1.
- (ii) Calculate, with its uncertainty, the internal
 1. diameter of the tube,
 2. volume of the tube.

[6]

- (b) (i) Distinguish between *precision* and *accuracy*.
- (ii) Two sets of readings, A and B, were obtained by two students in experiments to determine the acceleration due to gravity.

[All readings are in ms⁻².]

Set A: 7.88; 7.87; 7.86

Set B: 8.75; 10.23; 10.45

State whether or not the sets of values are precise and accurate.

[5]

Q32: JUN 2018 P3

- 1 (a) (i) Define the term *homogenous equation*.
- (ii) Explain why a homogenous equation is not always correct.
- (iii) The volume, V , of a liquid that flows through a pipe in time, t , is given by

$$\frac{V}{t} = \frac{\pi P r^6}{8 C l}$$

where P , is pressure difference between the ends of the pipe of radius, r , and length, l . C is a constant.

1. Determine the base units of C .
2. Hence suggest what the quantity C represents.

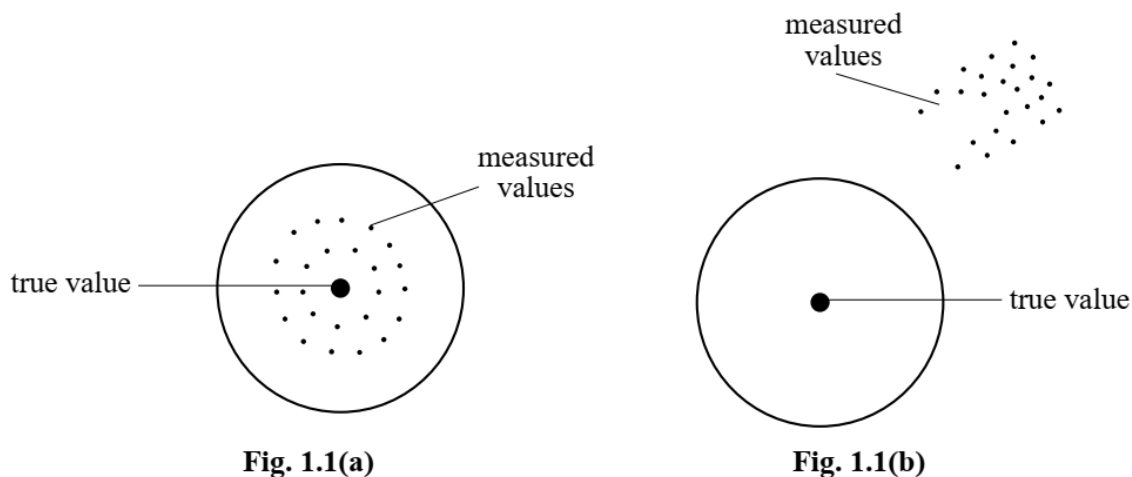
[6]

Q33: NOV 2018 P3

- 1 (a) (i) Express the *volt* in base units.
- (ii) Identify vectors and scalars from the following list of quantities:
- electric field strength*
tensile stress
momentum
- [4]
- (b) The resistive force, F , acting on a sphere depends on its radius, r , speed, v , and density, ρ , of medium.
- (i) State the base units of F , ρ , and v .
- (ii) Hence derive an equation for F in terms of ρ , v and r .
- [4]

Q34: 2018 SPEC P3

- 1 (a) (i) Define *systematic error*.
- (ii) List any **two** possible causes of random error.
- (iii) Fig. 1.1(a) and Fig. 1.1(b) shows the distribution of measurements taken by two students.



Comment on the precision and accuracy of the measurements.

[5]

Q35: JUN 2019 P3

- 1 (a) (i) State the numerical equivalence of
1. deci,
 2. giga.
- (ii) The gravitational potential, V , at a distance from the centre of the earth is given by $V = \frac{-GM}{r}$, where symbols have their usual meanings.

Show that the equation is homogeneous.

[4]

- (b) The velocity, v , of a wave on a string depends on tension, T , in the string, length, l and its mass, m .

Derive using base units, the equation connecting v in terms of T , l and m . [4]

Q36: NOV 2017 P3

Answer question 1 and any other question.

- 1 (a) In nuclear reactions the energy, E , and mass m , are related by the equation $E = mc^2$, where c is the speed of light.

- (i) Check the homogeneity of the equation.

- (ii) The period, T , of a simple pendulum of length l , is given by

$$T = 2\pi\sqrt{\frac{l}{g}}, \text{ where } g \text{ is the acceleration of free fall.}$$

Calculate the percentage change in T if l increases by 0.5%.

[4]